Final Interface & Evaluation Report

Jose Saravia

Link to document repository.

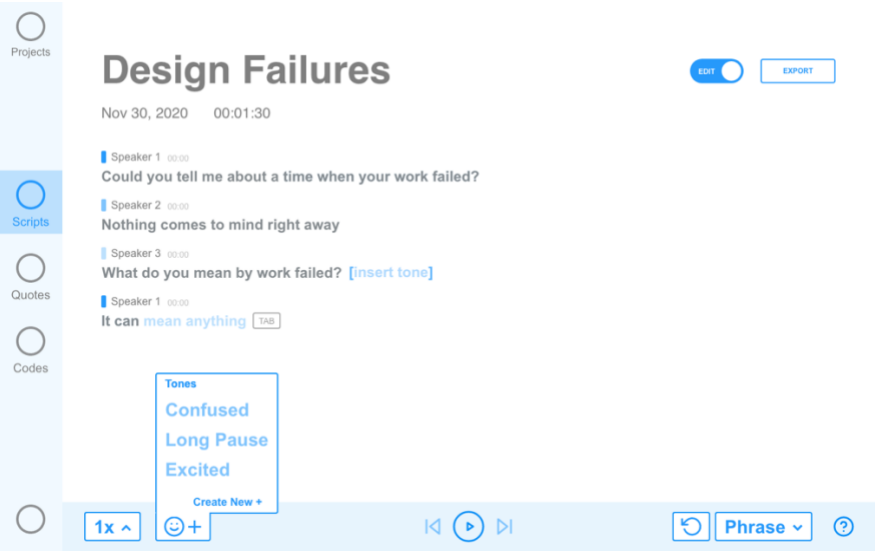
Functional Mid-Fi Prototype

A medium fidelity prototype was created to test more advance task that are difficult to mimic or explain using a low fidelity paper-prototype.

The following list describes the task that was tested in more detail. The descriptions below were not shown to any users testing the application. Instead, a short brief was shown to users.

- Use active assistive transcription to transcribe the audio file provided. This task requires users to use the transcription phrase suggestions to transcribe the audio. This feature uses keyboard commands and audio which was difficult to mimic on a paper prototype.
- Use tone tags to apply tonal language to the transcribed audio. This task requires users to apply a tone to a line mentioned by the speaker.
- Use audio measures to rewind the paused audio by 3 words. This task requires users to understand audio measures and change the settings from a phrase (default) to a word. Rewinding audio and audio chunking were difficult to mimic on a paper prototype.

Mid-Fi tests were conducted by sending participants the following link. Participants shared at least one of the following: college-educated or familiar with software design/development. A remote test allowed me to test the application without any potential intervention.



Inserting tonal language using drop-down menu. LINK TO MID-FI PROTOTYPE

Evaluation

Mid-Fi Prototype Analysis

After conducting multiple remote Mid-Fi Prototype test and reviewing the test videos, I discovered these 3 usability concerns.

Rewinding Audio by measure

- Description: Users had a difficult time knowing if they had clicked enough times to rewind the audio the appropriate measure amount. Unknowingly some users would quickly over-rewind.
- Heuristic: Lack of visibility of system status -> User doesn't know the number of measures it has reversed. Lack of recognition -> User should not need to remember the number of clicks.
- Solution: Show the user the number of times the rewind button has been clicked. This could be displayed as a counter bubble on the top right corner of the rewind button.

Discovering suggestion keyboard shortcut

- Description: Users would initially click on tab instead of press on the tab key. This could be a prototyping limitation (sometimes keyboard shortcuts interfere with browser shortcuts) or suggest a misunderstanding on how suggestions function.
- Heuristics: Flexible and efficiency of use -> Poor discoverability for suggestion shortcuts that would speed up the use of the system.
- Solution: 3 possible solutions: Make the keyboard shortcut more discoverable or attention grabbing. Allow the user two methods for selecting a suggestion (mouse click and key shortcut). Create an onboarding tutorial that introduces the controls to the user.

Understanding how audio measure work

- Description: Users seem to understand what an audio measure is but are confused by how its calculated by the system. Especially the concept of a phrase measure.
- Heuristic: Lack of help and documentation -> A user should know how the data inputted is being analyzed.
- Solution: This is less of a usability issue and, instead, an information issue. Properly onboarding and providing a help button could help inform the user of the system.

Static Hi-Fi Prototype Heuristic Review (Static Hi-Fi Prototype)

A high-fidelity static prototype was created to explore the graphic user interface design of the application. Features, layouts, and design choices were fleshed out. Afterward, I had designers critique the design implementation of my application: These are the key insights and advice I received:

- Easily identify speakers: A user would want to be able to easily distinguish who spoke a certain line of text. Don't make the user work or have a difficult time discovering speakers.
- Ability to replay measure: What happens if a user has reminded the audio and missed what the audio said.
- Inline tonal language editing: Users who are transcribing text don't want to use their mouse too often, so instead of using a dropdown menu to select tone considered an inline option.
- Reduce information displayed: The user might not need to always know all of the transcripts metadata (length, time, number of codes, etc.). Consider hiding some of it!
- Displaying codes: Showing all the code could clutter the interface, consider hiding or reducing the information shown.
- Ability to undo actions: A basic usability heuristic that should be followed.

Final Report

Problem

After collecting qualitative research data, researchers must clean and processes the data for analysis. This requires users to transcribe audio data, select the important text, and label selected text. This process is crucial but slow and repetitive. Processing data is a time-intensive process that prevents researchers from spending more time analyzing their data. There are three main roles researchers would perform: transcriber, reader, and organizer. The following task would be performed by each user role:

- Transcriber: Listen to audio files, transcribe audio into text, manipulate audio,
- Reader: Proofread transcription, select quotes, track conversation
- Organizer: Group quotes, apply codes to quotes, manage and clean codes

Design (Link to Prototype)

The final design implements the three major task: actively transcribing audio, selecting important quotes, and adding codes to a quote. These tasks are supported at a high level which means that detailed sub-task (ex: labeling speakers, adding tonal language, code suggestions) are not supported by the designed prototype. This prototype has a high-fidelity visual feal, yet mid-fidelity interactive feel. Important features such as audio playback and text selection have been implemented. The following design decisions were motivated by interface evaluations:

- **How a transcript is organized**: At the end of the day, users are generating content in the form of a transcript; therefore, how this content is organized is very important. Conducting paper prototypes test revealed that users become confused if the speakers are not labeled.

After fixing this issue, the heuristic evaluation revealed that the speaker and timestamp should be easily discoverable by the user. A user should be able to easily map a block of text to a speaker. Different ways of labeling speakers were considered (Image F). The final design included a color labeled speaker with an animated timestamp on hover (reduce information noise).

- **How users are shown quotes and tags**: The paper prototype test revealed that the flow for selecting text and adding a code did not confuse users. This is not a surprise, since this flow is very standard. Yet, the heuristic evaluations revealed some potential information organization and noise issues. Displaying all codes to the user created a cluttered interface and removed focus from the transcript (main content). To solve this, the ability to hide/show tags was added which allows users to choose how much information is displayed (Image E). This significantly cleaned the interface and reintroduced white space.
- **Information hierarchy:** Heuristic evaluations and discussions with designers revealed that users might be shown too much information. The application needs to allow users to focus on transcribing and organizing textual quantitative data. This requires a clean interface that hides rarely needed metadata. To solve this information such as word count, code count, and timestamps were hidden from the user under sub-menus. Additionally, as mentioned previously, codes can be hidden if the user finds them distracting. Reorganizing the information displayed on the main layout helped solve these information hierarchy issues (Image G).

Implementation: The final prototype was implemented using ProtoPie, a high-fidelity prototyping tool that allows designers to create highly interactive prototypes with minimal amounts of code. The implementation only focuses on front-end usability and no back-end implementation.

Evaluation:

Next steps: I would like to conduct remote user testing on the final prototype. The final prototype provides a high degree of interactive fidelity for certain features. It would be interesting to discover new usability concerns that are introduced by providing testers a high-fidelity prototype.

Some Unsolved problems: Inline tonal language could be solved by introducing a tooltip toolbar that appears when the tone is being added. Introducing audio measures to the user could be solved by creating a proper onboarding experience.

Reflection

I wish I had made some time to conduct proper qualitative user interviews regarding the pain points users experience when processing qualitative data. The problem was chosen based on my own experience and informal conversation with others who have processed qualitative data. A longer research phase might have revealed problems I had never considered. Additionally, I would have liked to have spent more time prototyping some of the more complex solutions I had envisioned in the project proposal. These include adaptive precision highlighting, advanced keyboard shortcuts, and suggested codes.

Appendix

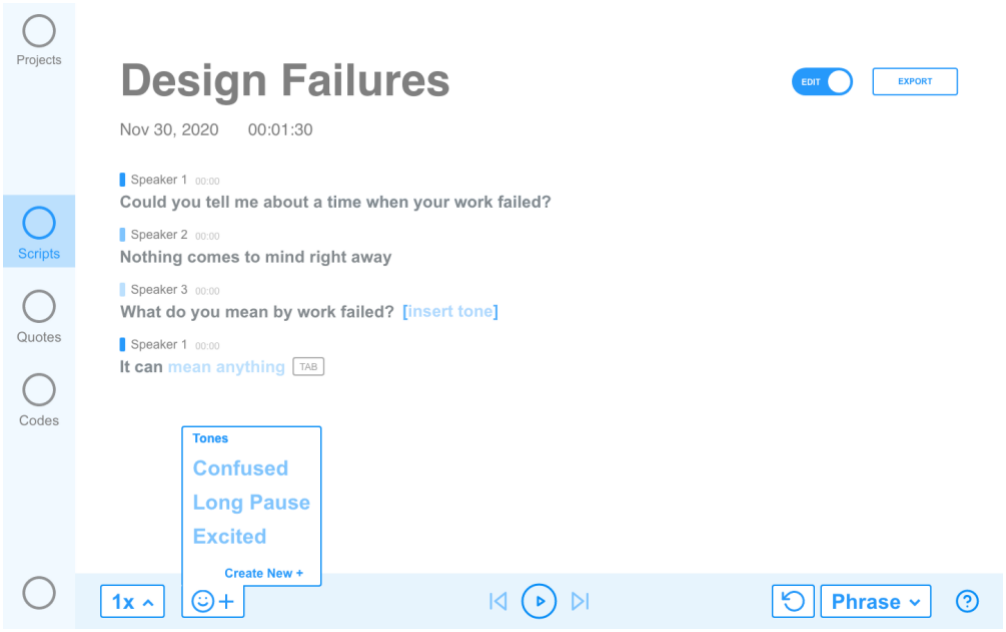


Image A: Mid-Fi mockup of selecting tonal language

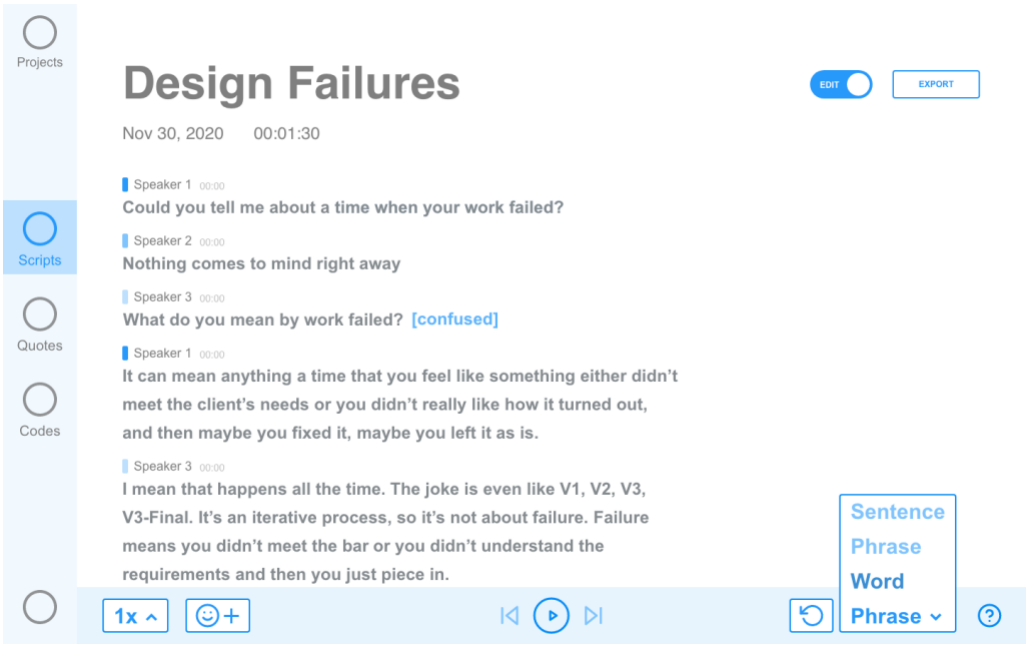


Image B: Mid-Fi mockup of selecting audio measure

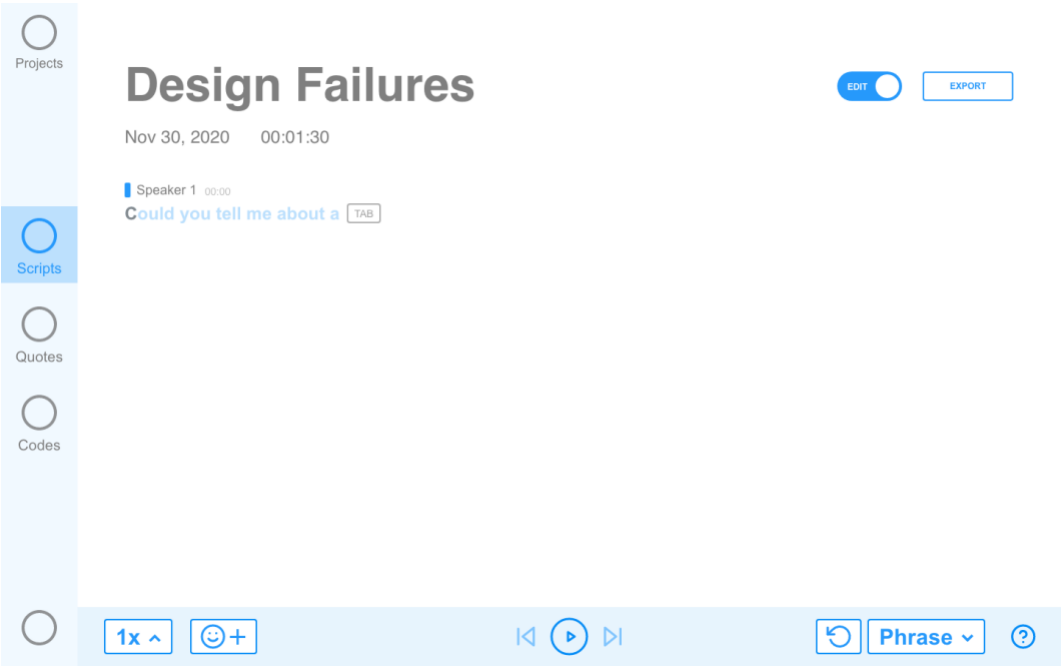


Image C: Mid-Fi mockup of transcription suggestion

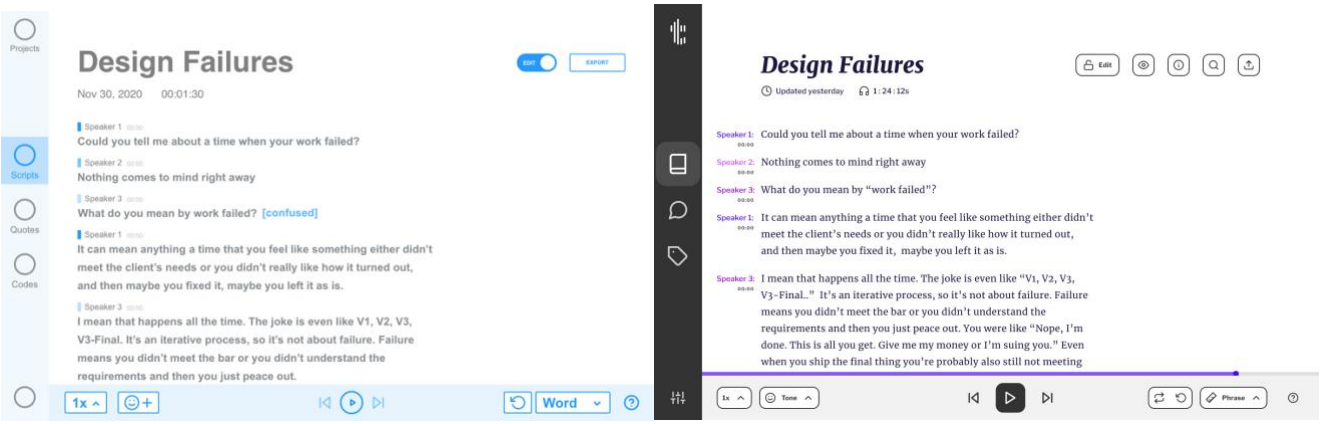


Image D: Mid-Fi to Hi-Fi prototype

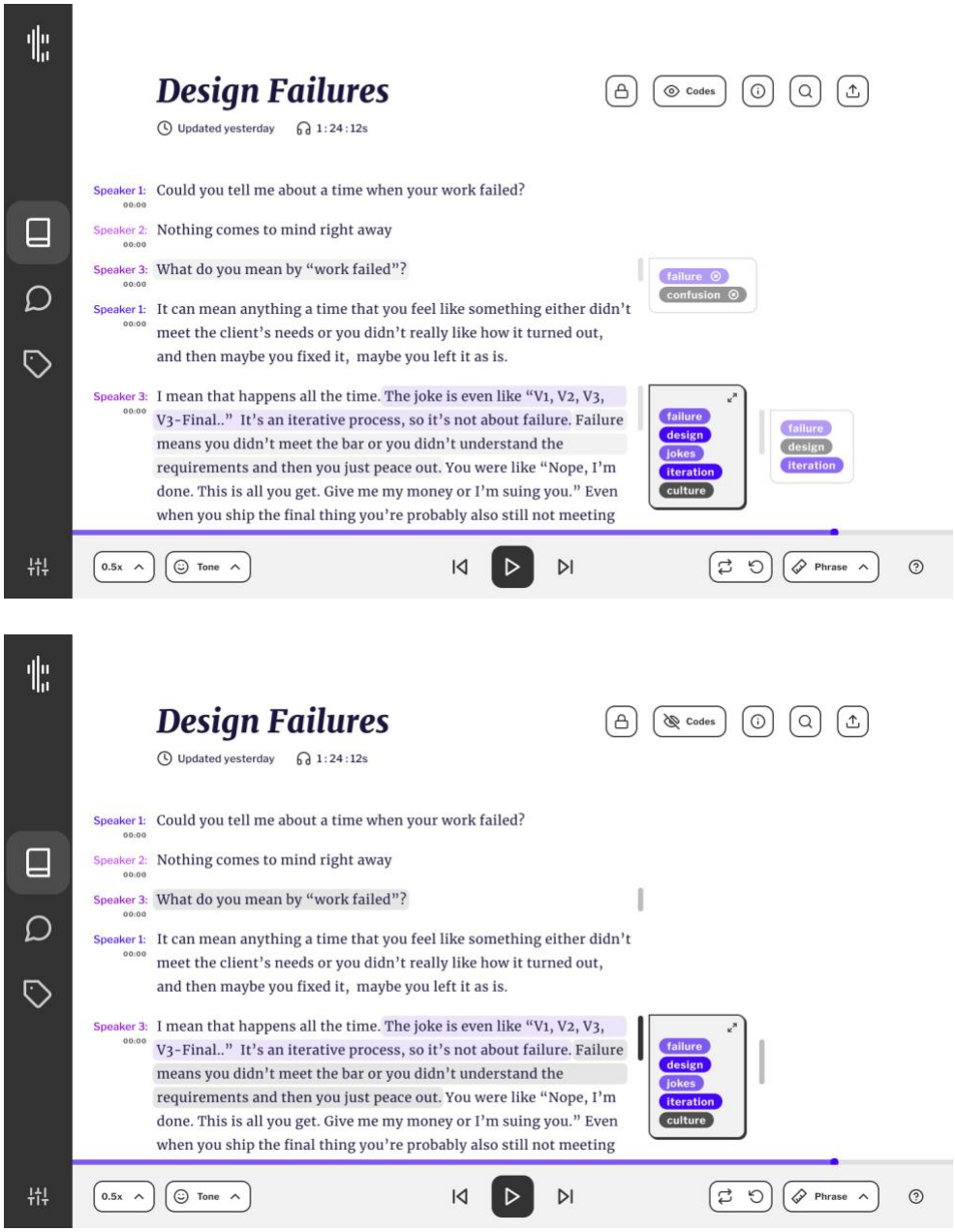
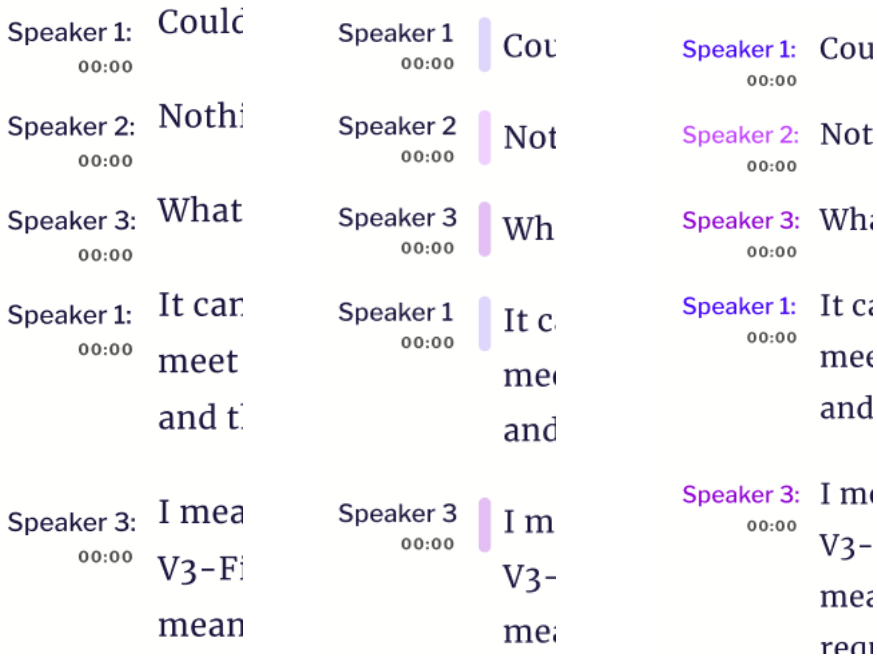


Image E: Cleaning up UI. The following changes were made: Introduced ability to hide/show tags, increased quote visibility, and increased selected quote emphasis.



V1 V2 V3-Final

Image F: Different iterations of organizing and displaying transcript speakers.

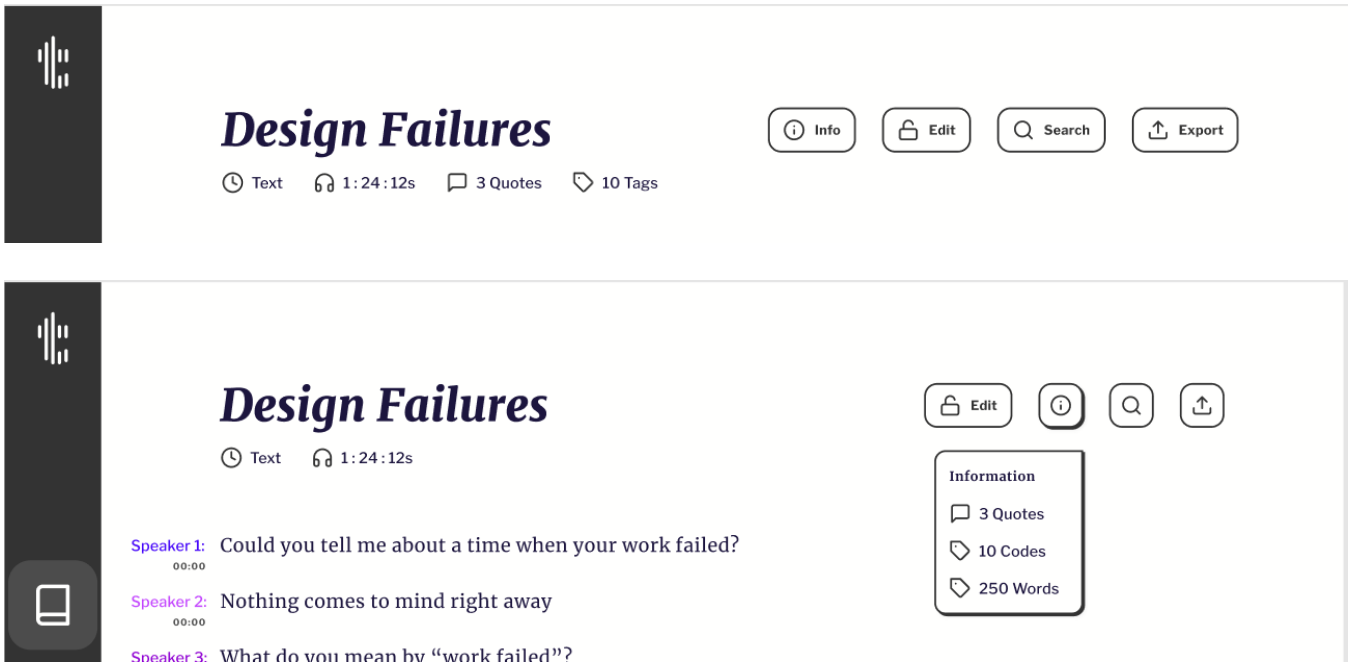


Image G: Reorganized how metadata is organized by the application. Metadata is hidden from the main interface.